

Prerequisites for other Linux distributions are mentioned in the user manual, which you can download from the link <http://sourceforge.net/projects/linkplustest/files/documentation/>.

You can now install LinK+ IDE using either one of the following methods.

Method 1

Download the LinK+ IDE installer from the link <https://sourceforge.net/projects/linkplustest/files/installers/>.

```
dileep@dileepk:~$ tar -xvf Link+\ IDE-linux.gtk.x86_64.tar.gz
dileep@dileepk:~$ cd Link+\ IDE-linux.gtk.x86_64/
dileep@dileepk:~/Link+\ IDE-linux.gtk.x86_64$ ./linkplus &
```

Method 2

Install LinK+ plugins. Before installing them, however, you should have a version of Eclipse CPP of the Helios variant already installed on your PC.

Start the Eclipse IDE and add the LinK+ repository by clicking on *Help->Install New Software*.

Then enter the URL of the updated LinK+ site (<https://sourceforge.net/projects/linkplustest/files/repository/>) and proceed with further installation procedures.

Method 3: Installation through Eclipse Market Place

Start the Eclipse IDE and click on *Help->Eclipse Marketplace*. Then, go to the *Search* button and in the *Find* space bar, type *Linux Kernel Programming IDE*. You can then click on the *Install* button to download the LinK+ plugins.

Now that you have installed the LinK+ IDE, we can start building our own projects.

The Linux kernel configuration and compilation project

Let us begin with configuring and compiling the Linux kernel, by selecting *File->New->Project*.

Click on the *Linux Kernel Development (Link+)* option and expand it to select *Kernel Compilation Project* before clicking on the *Next* button. This will open the kernel compilation wizard as shown in Figure 1, where we can enter the project's name. Then we need to select the Linux kernel source. Three options are provided: *Select Archived Source Code*, *Import Source Code to Workspace* and *Link to Existing Kernel Source Code*. Select one based on your requirement.

Depending on what Linux kernel source was selected, choose the path of the kernel source by clicking on the *Browse* button.

Next, move to the *Project Build* options. The LinK+ IDE is bundled with different project build options like *Configuration*, *Compilation* and *Installation*. These options are presented to users through multiple ways. Just right-click on the respective project in the *Project Explorer* view, then go

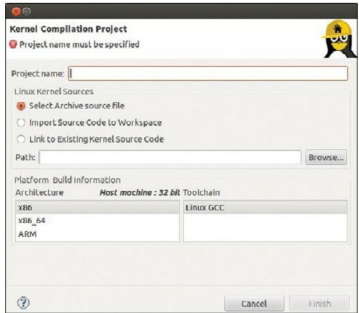


Figure 1: Kernel compilation project

to *Link* to or just click on the icon of the toolbar.

In my experience, how long compilation takes depends on your host system's hardware configuration. In order to speed things up, integrate third party tools such as *distcc* and *ccache* into this IDE. To use these features in the LinK+ IDE, go to the preference page (*Windows->Preference*) and go to *Linux Kernel Development*. If you want more information on *distcc* and *ccache*, refer to the user manual as suggested earlier.

Single step execution

The LinK+ IDE also provides a one-stop solution for all configuration, compilation, installation and emulation needs. To use this feature, right-click on the respective projects in the *Project Explorer* view, then go to *Link* to *-> Single Step Execution*.

This will open a wizard with three pages. The first page has configuration options, the second has compilation options and the third has the option to install on host or emulate with the QEMU file system of your kernel source.

Linux kernel emulation using QEMU

After compiling the Linux kernel image, the QEMU emulator can be used to emulate the working of the Linux kernel before installing on the host. For this process, there are two options—Linux kernel emulation using QEMU with the file system and with the *initrd* image. You can download the Debian or RPM based 'qcow2 file system' image from the Internet and use it for the emulation. Just right click on the respective project under *Project Explorer* and go to *Link* to *-> Emulation -> With File System*

The QEMU remote system

LinK+ IDE provides communication with remote systems using the *Eclipse Remote System Plugin*. The advantage of